

**BANARAS HINDU UNIVERSITY**

B.Sc. (Hons) Semester IV Examination 2025-26 (NEP)

**CHEMISTRY****Paper Code: CHEMJ43 : Physical Chemistry**

Ref. No.: Conf/Even/N/2025-26/07

Time: 2½ Hours

Maximum Marks: 70

*Instructions: Answer five questions including Question 1 which is compulsory. The figures in the right-hand margin indicate marks.*

**Question 1.**

- (a) Define partial molar volume. For a particular composition, explain how the negative, positive and zero partial molar volumes could be identified from the graph representing the variation of partial molar volume. [2]
- (b) What are Raoult's law and Henry's law? How can the ideal solution be defined based on these laws? [2]
- (c) Explain any two deactivation processes of a photochemically excited molecule. [2]
- (d) Write the equation explaining the effect of the inhibitor on the kinetics of the unimolecular surface reaction and explain, if the inhibitor is strongly adsorbed, then how the rate of reaction varies. [2]
- (e) Write briefly about the types of radioactive decay with suitable examples. [2]
- (f) During the use of radioisotopes, what are the points to be taken into account? [2]
- (g) What do you understand by the purity and strength of radioisotopes? [2]

**Question 2.**

- (a) Define fugacity and fugacity coefficient. When real gas behaves like an ideal gas, what would be the value of the fugacity coefficient? [3]
- (b) Write the expressions of Gibbs' energy of mixing and entropy of mixing and from these expressions, explain when and how the mixing of ideal gases is spontaneous. [4]
- (c) What is meant by chemical potential? How does the chemical potential vary with temperature and pressure? Derive the Gibbs-Duhem equation. [7]

**Question 3.**

- (a) Write the BET equation and explain how the graph needs to be plotted to obtain a straight line. [3]
- (b) What do you mean by (i) Quantum yield and (ii) Law of photochemical equivalence? [4]
- (c) Starting from the principle that at equilibrium the chemical potential of the pure solvent vapour equals the chemical potential of the solvent in the liquid mixture, derive the mathematical expression for the elevation of boiling point:  $\Delta T = Kx_B$ . State the assumptions made regarding the solute during this derivation. [7]

**Question 4.**

- (a) Define a photostationary state and explain how it differs fundamentally from a true chemical thermal equilibrium. [3]
- (b) Write the rate expression of a unimolecular surface reaction, sketching the graph for the surface covered vs. pressure and explain how the reaction rate varies from first-order at very low pressures to zero-order at extremely high pressures. [4]
- (c) Considering a suitable example of fluorescence quenching, derive the Stern-Volmer equation and find out the slope and intercept from the graph between  $I_0/I_f$  and  $[Q]$ . [7]

**Question 5.**

- (a) Discuss the influence of pH and temperature, depicting the graph, on the kinetics of enzyme-catalysed reactions. [3]
- (b) Discuss the photochemical decomposition of HI and find out its quantum yield. [4]
- (c) Using the steady-state principle for an intermediate enzyme-substrate complex (ES), derive the Michaelis-Menten equation. Draw the Lineweaver-Burk plot and explain what value could be obtained from its slope and intercept. [7]

**Question 6.**

- (a) Explain, with examples, transfer and reverse transfer reactions. [3]
- (b) Calculate the excitation energy of the compound nucleus  $^{26}\text{Al}^*$  formed by the bombardment of  $^{24}\text{Mg}$  with an 8.0 MeV deuteron ( $d$ ). [Given:  $m(^{24}\text{Mg}) = 23.9850 \text{ u}$ ,  $m(d) = 2.0141 \text{ u}$ ,  $m(^{26}\text{Al}) = 25.9869 \text{ u}$  and K.E. of projectile and target on Center of Mass System (CMS) = 7.38 MeV]. [3]
- (c) What is artificial radioactivity? Describe the experimental setup and evidence used by Frédéric and Irène Joliot-Curie to demonstrate the discovery of artificial radioactivity. [4]
- (d) Prove that the kinetic energy available for excitation of the target in the Lab System (LS) is equal to the total kinetic energy of the projectile and target in the CMS. [4]

**Question 7.**

- (a) How can one determine the structure of  $\text{PCl}_5$  using radioisotopes as tracers? [3]
- (b) Explain the principle of Isotope Dilution Analysis for determining the total blood volume in a human body. [3]
- (c) Define the nuclear reaction cross-section. Derive the relationship between the cross-section and the reaction rate for a beam of particles incident on a target. [4]
- (d) What is a Fricke dosimeter? Explain the detailed mechanism of the radiolysis in the Fricke dosimeter aqueous solution. [4]

**Question 8.** Answer any two of the following: [2 × 7 = 14]

- (a) Write the different steps for the photochemical reaction between  $\text{H}_2$  and  $\text{Br}_2$ , and obtain the expression for the rate of formation of  $\text{HBr}$ .

- (b) What is Osmometry? Find out the osmotic pressure at 25 °C of a solution containing 1 g glucose (M.Wt. 180 g/mole) and 1 g sucrose (M.Wt. 342 g/mole) per litre.
- (c) Write about Compton Scattering.

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